PG No. 20 D_{3d} $\bar{3}m$ (-3m1 setting) [trigonal]

* generator : $3^{+}_{001},\ 2_{110},\ -1$
* conjugacy class

$[1]:$ 1
$[2_{100}]:$ $2_{100},\ 2_{010},\ 2_{110}$
$[3^{+}_{001}]:$ $3^{+}_{001},\ 3^{-}_{001}$
$[-1]:$ -1
$[\mathrm{m}_{100}]:$ $\mathrm{m}_{100},\ \mathrm{m}_{010},\ \mathrm{m}_{110}$
$[-3^{+}_{001}]:$ $-3^{+}_{001},\ -3^{-}_{001}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	1	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
2	3^{+}_{001}	$\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
3	3^{-}_{001}	$\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
4	2_{110}	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
5	2_{100}	$\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
6	2_{010}	$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
7	-1	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
8	-3^{+}_{001}	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
9	-3^{-}_{001}	$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
10	m_{110}	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
11	m_{100}	$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1

continued ...

Table 1

No.	tag	matrix (polar)	det
12	m_{010}	$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1